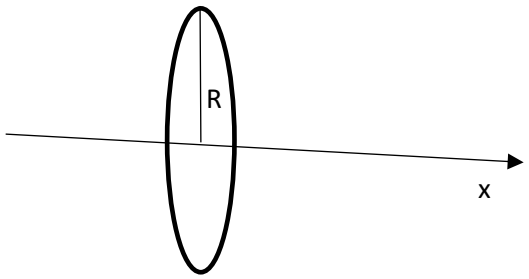
Exercise 1.



A circular current loop with radius *R* is oriented perpendicular to the *x*-axis as shown in the figure. Let $x = 0$ in the plane of the loop. A current circles counterclockwise in the loop, as seen from the positive half of the *x*-aksen. All magnetic fields other than that generated by the loop can be neglected. Which of the following statements about the magnetic field vector on the *x*-axis are correct?

A) The magnetic field points toward the right (along the *x*-axis) everywhere on the *x*-axis.

B) The magnetic field points to the right for positive values of *x* and to the left for negative values of *x*.

C) The magnetic field on the *x*-axis is perpendicular to this axis.

D) The magnitude of the magnetic field vector at $x = 0$ is zero.

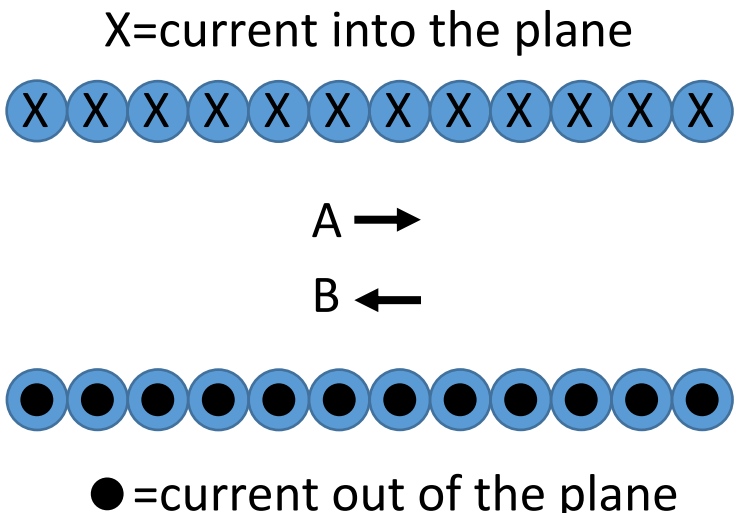
E) The magnitude of the magnetic field vector at $x = 0$ is proportional to $1/R$.

F) The magnitude of the magnetic field vector for $|x| \gg R$ is proportional to $1/x^2$.

G) None of the above statements are correct.

H) Don't know.

Exercise 2.



A proton having a magnetic dipole moment $\mu_p = 1.41 \cdot 10^{-26}$ J/T is situated in the middle of a long coil, of which a section is shown in the figure, cut through the middle in the longitudinal direction. The coil consists of a wires such that it has 500 turns per unit length. A current I=12.0 A is run through the coil in the direction shown in the figure. What is the difference in magnetic potential energy between the two orientations of the proton dipole, A og B, shown in the figure?

A) $E_A - E_B = 2.13 \cdot 10^{-28}$ J

B) $E_A - E_B = 1.06 \cdot 10^{-28}$ J

C) $E_A - E_B = 5.32 \cdot 10^{-29}$ J

D) $E_A - E_B = 3.38 \cdot 10^{-29}$ J

E) $E_A - E_B = 0$ J

F) $E_A - E_B = -3.38 \cdot 10^{-29}$ J

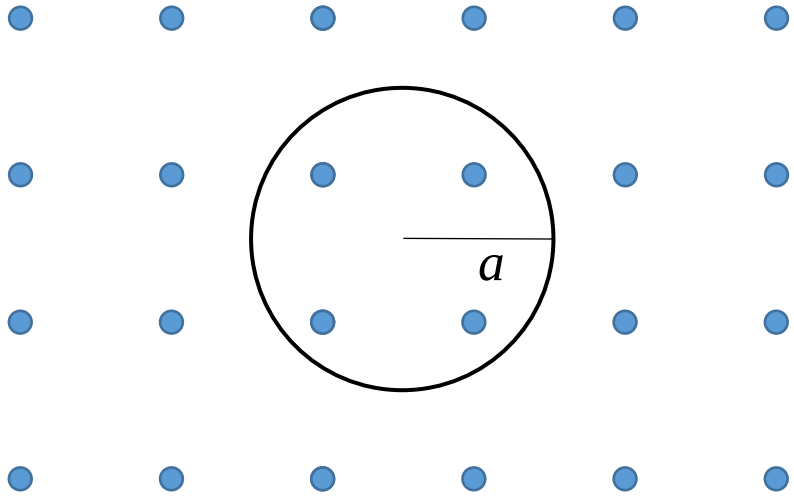
G) $E_A - E_B = -5.32 \cdot 10^{-29}$ J

H) $E_A - E_B = -1.06 \cdot 10^{-28}$ J

I) $E_A - E_B = -2.13 \cdot 10^{-28}$ J

J) Don’t know

Exercise 3.

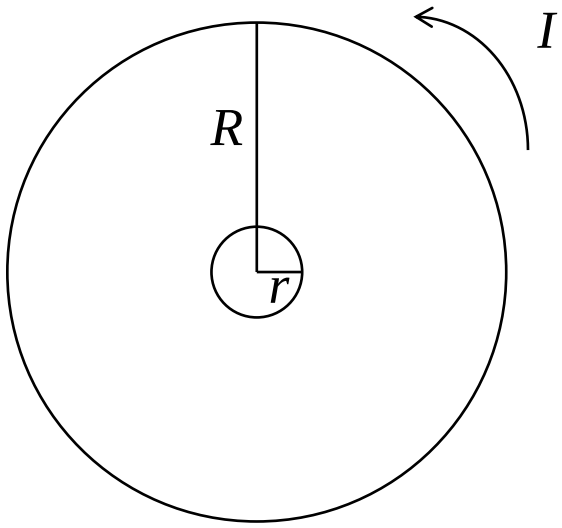


A circular conductor having a radius a = 20.0 cm is situated in a homogeneous magnetic field pointing out of the plane of the figure. The total resistance of the conductor is R_0=10.0 Ω.

a) Determine the direction of the induced current if the magnetic field strength increases or decreases.

b) Assume that the magnetic field is 0 for t<0, and grows as $B(t) = B_0\left(1 - e^{-\frac{t}{t_0}}\right)$ at later times. Derive an expression for the current in the conductor as a function of time, and plot the function for B_0=0.1 T, t_0=2.00 ms.

Exercise 4.



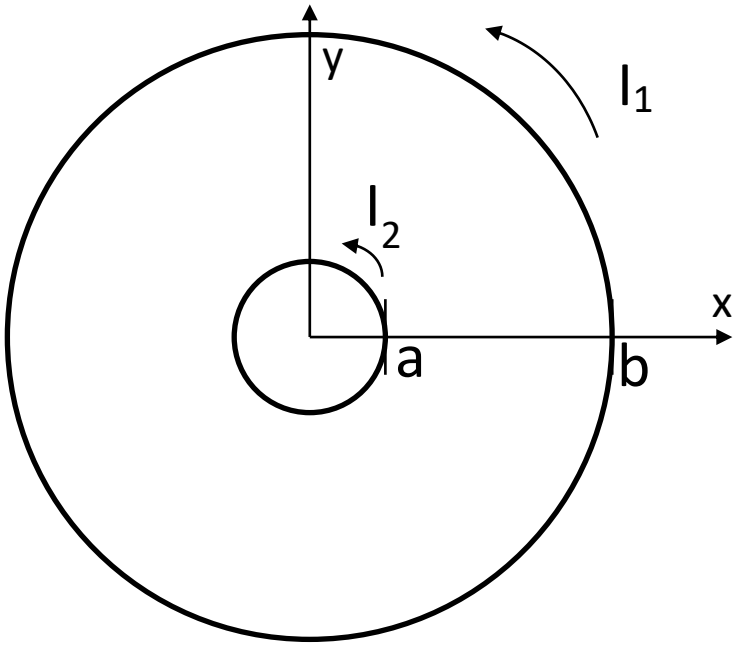
A circular current loop of radius *R* carries a current $I(t) = I_0 \cos(\omega t)$ which is taken to be positive in the counterclockwise direction shown in the figure. In the center of the loop another circular conductor is situated with radius $r << R$. We assume that the magnetic field from the outer loop can be taken constant over the cross section of the inner loop. The inner loop has a resistance R_0.

a) Find an expression for the induced emf in the inner loop.

b) Find an expression for the current in the inner loop. Plot it as a function of time, together with *I(t)* for R=1.00 m, r=1.00 cm, I_0=10 A, $\omega = 2\pi \cdot 50\ s^{-1}$ (50 Hz frequency), R_0=0.1 Ω. Neglect in this calculation the magnetic field created by the current in the inner loop.

c) What is the magnitude at the center of the loops of the magnetic field created by the induced current in the inner loop? How good was the approximation in b)? Could there be cases where it does not hold at all?

Exercise 5.



Two concentric current loops centered on origo have radii $a = 1.00$ cm, $b = 2.00$ m, and carries the currents $I_1 = 2.00$ A and $I_2 = 5.00$ A as shown in the figure (for clarity, they are not drawn to scale). As the inner loop is very small relative to the outer one, it can be assumed that the field from the outer loop is constant over the cross-section of the inner loop, *i.e.* it is identical to the field of the outer loop in the center. Assuming the outer loop to be fixed, determine the W that must be done on the inner loop to rotate it 180 degrees around the *y*-axis.

A) $W = -1.97$ nJ

B) $W = -0.987$ nJ

C) $W = -0.628$ nJ

D) $W = 0.628$ nJ

E) $W = 0.987$ nJ

F) $W = 1.97$ nJ

G) Don't know